

HARSH SHROFF

AI/ML Engineer — Open to Relocation

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PROFESSIONAL SUMMARY

AI/ML Engineer specializing in architecting **Neuro-Symbolic AI Agents** and production-grade RAG systems for regulated industries. Expert in multi-model orchestration and building **verifiable agentic workflows** that transition legacy manual processes into automated reasoning engines. Proven track record of achieving **60% efficiency gains** in document automation and deploying high-accuracy CV pipelines, with a core focus on **AI safety, system observability, and scalable MLOps**.

CORE COMPETENCIES

Machine Learning & AI: Generative AI, LLM Agents, Neuro-Symbolic AI, **PydanticAI**, LangChain, RAG, Prompt Engineering, PyTorch, TensorFlow, Computer Vision, YOLOv8, OpenCV

Deep Learning Specializations: Neural Networks, CNNs, RNNs, Transformers, Fine-tuning, RLHF, Anomaly Detection, Reinforcement Learning, Time Series

Software & Infrastructure: Python, FastAPI, SQL, **Docker**, CI/CD, **AWS** (Bedrock, SageMaker, Lambda, Textract), MLflow, Vector Databases (Pinecone/pgvector), RESTful APIs

Cloud & MLOps: AWS (EC2, S3), MLflow, Weights & Biases, Model Monitoring, Model Governance, A/B Testing, Auditability

Data Engineering: Pandas, NumPy, Spark, Hadoop, Feature Engineering, Data Pipelines, ETL, PostgreSQL, MongoDB, Redis

Deployment & Tools: Streamlit, Gradio, Jupyter, CUDA, Edge Computing, NVIDIA Jetson, Model Serving, Performance Optimization

PROFESSIONAL EXPERIENCE

AI/ML Researcher

Mar 2023 – Present

UMBC Center for Real-time Distributed Sensing and Autonomy

- Developed distributed ML systems for multi-modal sensor data fusion under US Army Research Lab funding, achieving **20% performance improvement** in real-time optimization systems. *Python, PyTorch, TensorFlow*
- Built **CHARLIE Voice Assistant**, a voice-enabled AI agent integrating speech recognition, LLM inference, and real-time dialogue management for autonomous system interfaces. *Voice AI, LLM Integration, Speech Processing*
- Engineered production-ready multi-modal ML systems integrating advanced perception algorithms for next-generation AI applications using deep learning frameworks and edge computing platforms. *Computer Vision, NVIDIA Jetson*

AI/ML Engineer - Production AI Systems

Jun 2024 – May 2025

VITG Corp., Halethorpe, MD

- Architected a production-grade **RAG automation system** using AWS Bedrock and SageMaker (Claude 3.5), reducing candidate screening time by **60%** and processing 2,500+ regulatory documents with 40% higher efficiency.
- Implemented **semantic search architecture** using vector embeddings and AWS Textract for document ingestion, ensuring 100% auditability and SOC2-compliant logging for regulatory workflows.
- Established **ML governance frameworks** and monitoring pipelines for LLM observability, utilizing automated evaluation metrics to mitigate hallucinations and maintain production reliability.
- Developed scalable ETL pipelines and RESTful APIs for multi-model LLM inference, enabling seamless integration with enterprise systems and cross-functional team collaboration. *FastAPI, PostgreSQL, Docker, Multi-Model Orchestration*

Data Scientist Intern - Computer Vision

Aug 2023 – Dec 2023

The Conservation Fund, Shepherdstown, WV

- Developed production-ready **computer vision system** achieving **92% accuracy** using YOLOv8 and OpenCV, deployed on edge devices for real-time quality assessment with end-to-end ML pipeline from data collection to model serving. *Computer Vision, Edge Deployment, MLOps, Raspberry Pi*
- Co-authored peer-reviewed research demonstrating commercial impact of deep learning applications, enabling commercial deployment of AI quality control systems with automated reporting capabilities. *Research Publication, Production Deployment, Technical Writing*

WeHear Innovations Pvt. Ltd., Ahmedabad, India

- Developed **Personal Hearing Intelligence (PHI)** machine learning models, analyzing longitudinal audio exposure and ambient noise patterns to generate personalized hearing-risk metrics.
- Built **time-series feature engineering pipelines** for high-frequency sensor data, enabling early-risk detection through statistical monitoring of volume and frequency band exposure.
- Optimized **audio signal processing workflows** for bone-conduction hardware, implementing low-latency ML inference to support real-time voice amplification features.
- Collaborated with firmware teams to integrate model outputs into production mobile applications, translating raw audio predictions into user-facing clinical health insights.

KEY TECHNICAL PROJECTS

Enterprise Document Intelligence Platform

AWS Bedrock, SageMaker, Textract, S3, Lambda, Python, LLM APIs, Streamlit

- Built scalable document processing system automating extraction from 2,500+ PDFs using LLM-powered OCR pipelines with AWS Bedrock and SageMaker, implementing RAG architecture for intelligent document querying and automated compliance workflows.
- Designed prompt engineering strategies and model evaluation frameworks for multi-LLM orchestration, achieving high accuracy in regulatory document classification and information retrieval with governance-compliant audit trails.
- Implemented interactive dashboards enabling stakeholders to query and visualize insights from unstructured data with automated reporting, model performance monitoring, and decision-support capabilities.

Multi-Modal Recommendation Engine ([GitHub](#))

Python, OpenAI API, Computer Vision, PostgreSQL, A/B Testing

- Engineered hybrid recommendation system combining structured data analysis with computer vision and LLM-based personalization, reducing user bounce rate by **20%** and increasing engagement by **40%** through prompt-engineered AI-driven recommendations.
- Implemented scalable data pipelines and A/B testing framework for continuous model improvement and performance monitoring with real-time analytics and user feedback integration.

Bio-Oracle: Neuro-Symbolic Agentic AI ([GitHub](#))

Python, PydanticAI, Gemini, Cellpose, Docker, Robust Z-score Normalization

- Developed a **Neuro-Symbolic Agent** for high-throughput phenotypic screening, automating discovery processes on the BBBC021 dataset by combining deep learning segmentation with LLM-based reasoning.
- Built a custom **Feature Factory** to extract 30+ morphological features and implemented a Robust Z-score normalization pipeline, optimized for local inference on Apple Silicon (MPS).
- Architected the agentic workflow using **PydanticAI** to provide structured, verifiable reasoning over biological data, packaged in a production-ready Docker container with a focus on SEO and system auditability.

EDUCATION

Master of Science, Data Science

Aug 2022 – May 2024

University of Maryland Baltimore County (UMBC) — GPA: 3.8/4.0

- **Relevant Coursework:** Machine Learning, Deep Learning, NLP, Computer Vision, Big Data Systems, Statistical Analysis, Predictive Modeling

CERTIFICATIONS & PUBLICATIONS

Professional Certifications

- **AWS Machine Learning Associate Certification** – [View Credential](#) – Active through Apr 2028
- **AWS Machine Learning Foundations** – Core ML concepts, cloud deployment, and AWS service integrations
- **NVIDIA Deep Learning Institute** – Transformer-Based NLP Applications, GPU-Accelerated Computing

Publications & Research

- **Ranjan, R., Shroff, H., et al.** (2024). "FilletCam AI: Precision color profiling using deep learning." *Journal of Agriculture and Food Research*. DOI: [10.1016/j.jafr.2024.101461](https://doi.org/10.1016/j.jafr.2024.101461)
- **Trivedi, K., Shroff, H.** (2021). "Mosquito identification using ML on embedded systems." *IEEE ITU Kaleidoscope Conference*. DOI: [10.23919/ITUK53220.2021.9662116](https://doi.org/10.23919/ITUK53220.2021.9662116)